

LC 464: Can I Win

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Game Mechanics & Breakdown

basically in 100 game, 2 player take turn. player pick any number bw [1, 10]. whoever reaches 100 first wins.

GAME change:

- **100** → desired total
- **[1, 10]** → a max integer can be given, so **[1, max]**
- **(CANT repeat same number)**

so → when the player chooses a number it's added to running total, hence no individual score.

Learning of the day:

can't repeat same number, so first instinct of mine was to use a used vector but then...

`dp[i][used]` → used itself is a vector → so this was giving a error.

Solution: use **BITMASK** **[1, 0, 1, 0]** → **(1010)**

Core Logic & Intuition

- basic idea is if my move can make other player loose i win
- once i do my turn, i turn 0 bits into 1 bits in mask
- **HENCE larger masks are processed first as they are prerequisites**
- the answer will be given by `dp[0]` because 0 means nobody no choosen any number for now

C++ Solution Implementation

```
//AREE YOU WINNING SON????

class Solution {
public:
    bool canIWin(int maxchoosableinteger, int desiredtotal) {

        if(((maxchoosableinteger*(maxchoosableinteger+1))/2)<desiredtotal)
return false;
        int states=1<<(maxchoosableinteger+1); //+1 kyunki 1 indexed nums
he , no 0
        vector<int> dp(states,false); //filhaal koi nhi jeet skta , agar
meri move ke baad saamne waala har gaya
//to main jeet gaya

        //meri move ke baad 0 bit-> 1 bit , hence bada mask , to bada mask
pehle process krna hoga

        for(int mask=states-1;mask>=0;mask--){
            int currsum=0;

            for(int i=1;i<=maxchoosableinteger;i++){
                if(mask&(1<<i)){
                    currsum+=i;
                }
            }

            //running total mil gaya
            //ab kisi andekhe ko add krke dekhenge jeete ya ni
            for(int j=1;j<=maxchoosableinteger;j++){
                if(mask&(1<<j)){
                    continue;
                }

                if(currsum+j>=desiredtotal){
                    dp[mask]=true;
                    break;
                }

                int newmask=mask | (1<<j);
                if(dp[newmask]==false){ //saamne waala mere move ke baad
haar gaya
//to main jeeet gaya

                    dp[mask]=true;
                    break;
                }
            }
        }

        return dp[0];
    }
};
```

```
};  
}
```